

MATRIXPOLICY: ROLE-AWARE OPTIMIZATION FOR RATIONAL TRANSFORMER NONLINEARITIES

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ABSTRACT

Modern LLM development is constrained by how many pretraining tokens and hours are needed before a model is useful enough to enter the downstream loop of evaluation, post-training, and iteration. We make the nonlinear Transformer sublayer optimizer-visible. Rational Latent Basis (RLB) replaces a fixed response with grouped, RMS-normalized, real-pole-free P5/Q4 rational responses and exposes derivative gains, response gains, coefficient activity, and matrix-pair balance summaries. MatrixPolicy consumes these summaries to update only the adjacent input and output matrices, while rational coefficients and all other Transformer parameters remain on AdamW. On five pretraining corpora with a fixed 12-layer, width-768 decoder, MatrixPolicy reaches matched validation-loss targets with fewer tokens and lower measured arrival time than SiLU controls. It saves 19.5–22.8% target-arrival tokens versus SiLU with AdamW and 29.2–34.0% versus SiLU with Muon at 100M tokens, and 22.3–26.4% and 6.7–9.8%, respectively, at 300M tokens. [todo: Insert the larger-scale training result here.]

1 INTRODUCTION

Current LLM development is limited by the capability purchased per token, per optimizer step, and per hour of training, rather than by final model size alone. A pretraining run is not an isolated curve-fitting exercise: it gates architecture search, data-mixture revision, post-training, safety evaluation, and release cadence. Recent open model reports therefore foreground training data, token budgets, architecture choices, stability, and hardware cost alongside benchmark quality (Grattafiori et al., 2024; Qwen et al., 2025; DeepSeek-AI et al., 2024). Recent scaling and training-protocol studies make the same point from a more controlled angle: loss depends on data mixture, repetition, optimizer choice, and the full training configuration, so small design changes should be judged by how quickly they reach useful loss, not only by the endpoint after a fixed budget (Li et al., 2025; Liu et al., 2026; Volkova et al., 2026; Semenov et al., 2025; Wen et al., 2026). We therefore use target arrival as the main readout: when two runs eventually approach similar endpoint losses, the one that reaches the same validation loss earlier has changed the cost of pretraining.

The nonlinear Transformer sublayer is a plausible place to improve target arrival, but only if it is treated as a matrix interaction rather than as an isolated scalar curve. Early Transformer implementations used pointwise responses such as ReLU or GELU (Vaswani et al., 2017; Hendrycks & Gimpel, 2016). Modern decoder blocks instead often use GLU-derived gated sublayers, including SwiGLU in PaLM and LLaMA-style models (Dauphin et al., 2017; Shazeer, 2020; Chowdhery et al., 2022; Touvron et al., 2023). This made nonlinear sublayers stronger, but it also made many fixed-curve swaps less structurally decisive: learned projections choose latent coordinates, a response transforms them, and another matrix recombines the features. The scalar curve matters, but the surrounding matrices decide what the curve sees and how its features return to the residual stream.

Learned rational responses are a natural response family for this interaction. Rational neural networks have approximation-theoretic advantages for smooth functions, and recent theory extends the separation to modern fixed activations, including gated and Transformer-style nonlinearities (Boullé et al., 2020; Tang & Townsend, 2026). Prior Transformer work with rational activations shows that replacing a fixed curve by a trainable rational curve can help (Fang et al., 2023). The missing piece is the optimizer interface. If the rational curve is trained as just another tensor, the optimizer does not

know which adjacent matrix selects rational coordinates, which matrix recombines rational features, or which reciprocal rescalings leave the represented sublayer map nearly unchanged.

Rational Latent Basis (RLB) exposes exactly these quantities. It maps residual states into grouped latent coordinates, normalizes each group by its RMS radius, and applies a trainable real-pole-free P5/Q4 rational response. This parameterization produces response gains, derivative gains, coefficient-activity signals, relative matrix-pressure summaries, and a positive matrix-pair balance coordinate. MatrixPolicy is the optimizer rule paired with that interface: it updates only the RLB input and output matrices by role, while rational coefficients, attention weights, embeddings, normalization parameters, and all other Transformer tensors remain on AdamW.

The completed fixed-scale study uses the same 12-layer, width-768 causal Transformer across DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4. At the 100M-token budget, MatrixPolicy reaches shared validation-loss targets earlier than matched SiLU with AdamW and SiLU with Muon baselines. At the 300M-token budget, where longer runs compress endpoint gaps, the target-arrival advantage remains visible in both tokens and measured time. [todo: Insert the larger-scale training result here.] The contribution is an activation design that exposes optimizer-visible matrix roles, a localized role-aware optimizer for the exposed matrix pair, and matched evidence that this interface reduces target-arrival tokens and wall-clock time across five pretraining corpora.

2 RELATED WORK

The nonlinear sublayer is where scalar response design became entangled with learned matrix roles. Early Transformer-style sublayers used fixed pointwise responses such as GELU or Swish/SiLU (Vaswani et al., 2017; Hendrycks & Gimpel, 2016; Ramachandran et al., 2017). GLU introduced multiplicative feature selection for language modeling (Dauphin et al., 2017), GLU variants such as SwiGLU improved Transformer sublayers in compute-matched settings (Shazeer, 2020), and large decoder models adopted SwiGLU together with pre-normalized Transformer components (Chowdhery et al., 2022; Touvron et al., 2023). This history motivates treating the nonlinear block as a latent-matrix map, not as a scalar curve in isolation. RLB keeps that sublayer location but exposes the grouped coordinates, curve gains, and adjacent matrix roles that fixed-curve GLU variants leave implicit.

A rational response is the scalar side of this interface. Padé Activation Units showed that low-degree quotients can be learned end to end as activation functions (Molina et al., 2020), rational neural networks gave approximation efficiency results for smooth-function classes (Boullé et al., 2020), and RAFT evaluated rational activation functions inside Transformer pretraining and fine-tuning (Fang et al., 2023). Recent theory strengthens the motivation by proving expressivity advantages for trainable low-degree rational activations over a broad set of modern fixed activations, including gated and Transformer-style nonlinearities (Tang & Townsend, 2026). Our use of rational functions follows this family, but the empirical question is not whether a trainable curve alone is helpful. The controls distinguish a global-rational RLB activation without MatrixPolicy from the same activation paired with the optimizer-visible matrix interface.

The matrix side connects RLB to LLM pretraining optimizers. Adam and AdamW remain central baselines (Kingma & Ba, 2015; Loshchilov & Hutter, 2019), while state-efficient or diagonal adaptive methods such as Adafactor and CAME reduce optimizer memory or alter adaptive-state estimates (Shazeer & Stern, 2018; Luo et al., 2023). Other recent methods change the update rule itself: Lion, Schedule-Free methods, and cautious optimizers alter momentum, sign, or schedule behavior (Chen et al., 2023; Defazio et al., 2024; Liang et al., 2026), while Sophia, GaLore, and Adam-mini change curvature, low-rank, or per-block learning-rate structure (Liu et al., 2024; Zhao et al., 2024; Zhang et al., 2025). Matrix-aware optimizers are the closest context: Shampoo uses tensor preconditioners (Gupta et al., 2018), SOAP stabilizes Shampoo-style preconditioning by running Adam in the preconditioner eigenbasis (Vyas et al., 2025), and Muon uses orthogonalized matrix updates for LLM training (Liu et al., 2025). Recent optimizer comparisons emphasize fair tuning, model-scale dependence, and the fact that matrix-structured optimizers can change token or compute efficiency without uniformly dominating across settings (Zhao et al., 2025; Wen et al., 2026). MatrixPolicy follows the matrix-aware direction, but RLB supplies role information beyond tensor shape: derivative and response gains, relative input/output pressure, coefficient activity, and pair-balance signals applied only to the nonlinear sublayer’s input/output matrix pair.

3 METHOD

RLB and MatrixPolicy are designed as a single interface. RLB defines the nonlinear sublayer and the detached summaries it exposes; MatrixPolicy consumes those summaries to update the two adjacent matrices. Rational coefficients and ordinary Transformer parameters remain on the AdamW path in the evaluated configuration.

Throughout the method we use

$$\text{clip}(x, a, b) = \min\{\max\{x, a\}, b\}, \quad (1)$$

$$\text{smoothstep}(a, b, x) = 3\omega^2 - 2\omega^3, \quad \omega = \text{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \quad (2)$$

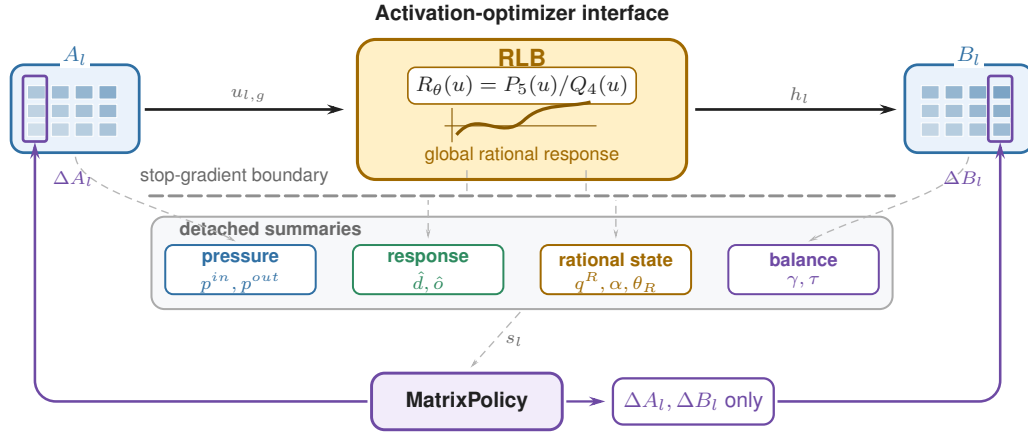


Figure 1: Activation–optimizer interface for global-rational RLB. In layer l , A_l maps the sublayer input into normalized rational groups and B_l recombines the rational features. RLB exposes matrix-pressure, response-gain, coefficient-activity, and pair-balance summaries through a stop-gradient boundary. MatrixPolicy reads these summaries and writes only ΔA_l and ΔB_l ; rational coefficients are AdamW-trained and policy-observed, not MatrixPolicy update targets.

3.1 RATIONAL LATENT BASIS

For layer $l \in \{1, \dots, L\}$, let $x_l \in \mathbb{R}^d$ be the input to the nonlinear sublayer. RLB uses an input matrix $A_l \in \mathbb{R}^{H \times d}$ and an output matrix $B_l \in \mathbb{R}^{d \times H}$ with latent width $H = Gm$. Let $A_{l,g} \in \mathbb{R}^{m \times d}$ denote the row block of A_l , and let $B_{l,g} \in \mathbb{R}^{d \times m}$ denote the matching column block of B_l . Each group has a trainable P5/Q4 rational response

$$P_{l,g}(u) = \sum_{i=0}^5 a_{l,g,i} u^i, \quad (3)$$

$$Q_{l,g}(u) = 1 + |b_{l,g,1}| |u| + |b_{l,g,2}| u^2 + |b_{l,g,3}| |u|^3 + |b_{l,g,4}| u^4, \quad (4)$$

$$R_{l,g}(u) = P_{l,g}(u)/Q_{l,g}(u). \quad (5)$$

The RLB forward map is the single block expression

$$y_l = \sum_{g=1}^G B_{l,g} \left[\rho_{l,g}(x_l) R_{l,g} \left(\frac{A_{l,g} x_l}{\rho_{l,g}(x_l)} \right) \right], \quad \rho_{l,g}(x_l) = \sqrt{m^{-1} \|A_{l,g} x_l\|_2^2 + \epsilon_{\text{rms}}}. \quad (6)$$

The scalar functions $P_{l,g}$, $Q_{l,g}$, and $R_{l,g}$ are applied coordinatewise when their input is a vector. Since $Q_{l,g}(u) \geq 1$ for all real u , the implemented response has no real pole. The response is piecewise rational because of the absolute-value terms; derivative statistics use the autodiff convention

sign(0) = 0 and equations involving $R'_{l,g}$ are interpreted almost everywhere. Numerator and denominator coefficients are initialized by a SiLU fit on $[-5, 5]$ and are then trained by AdamW. The evaluated model uses only these group-specific global P5/Q4 rational responses.

The forward map in equation 6 separates the normalized coordinate at which the rational response is evaluated from the group radius returned to the residual stream. The floor $\epsilon_{\text{rms}} > 0$ is an additive squared-RMS stabilizer. For fixed curves $R_l = \{R_{l,g}\}_{g=1}^G$, write this block map as $f_l(x; A_l, B_l, R_l)$. For any fixed input x such that $A_{l,g}x \neq 0$ for every group, if $\epsilon_{\text{rms}} = 0$, then for any $\lambda \in \mathbb{R}_{>0}^G$ and $D_l(\lambda) = \text{diag}(\lambda_1 I_m, \dots, \lambda_G I_m)$,

$$f_l(x; A_l, B_l, R_l) = f_l(x; D_l(\lambda)A_l, B_l D_l(\lambda)^{-1}, R_l). \quad (7)$$

This positive matrix-pair rescaling is exact only in the unfloored map. With the stabilizing RMS floor and optimizer state, the same direction becomes a bounded balance coordinate for MatrixPolicy rather than an exact invariance of the full training dynamics.

3.2 MATRIXPOLICY

MatrixPolicy acts only on the RLB matrix pair (A_l, B_l) . The RLB calculus exposes the signals used below: live response/derivative gains for output/input roles, relative matrix-gradient pressure, coefficient activity, and a positive pair-balance coordinate. After backpropagation at step t , MatrixPolicy forms bounded group/role multipliers, applies the AdamW/Muon matrix branch to the scaled gradients, and then applies the scheduled positive pair-rescaling move. Gradient statistics are computed before parameter updates; the pair-rescaling ratio is computed after the AdamW/Muon branch. For any tensor V , define

$$\text{rms}(V) = \sqrt{\text{mean}(V^2)}, \quad \text{rms}_\epsilon(V) = \sqrt{\text{mean}(V^2) + \epsilon}. \quad (8)$$

The constants $\epsilon_p > 0$ and $\epsilon_g > 0$ denote fixed additive floors in squared-RMS units. The same ϵ_p is reused as the implementation floor for matrix-ratio logs in equation 27. The resulting pressure and activity scores are bounded update-control signals. Let $A_{l,g} \in \mathbb{R}^{m \times d}$ be the row block of A_l , and let $B_{l,g} \in \mathbb{R}^{d \times m}$ be the column block of B_l . The relative matrix-gradient scales are

$$p_{l,g}^{\text{in}} = \frac{\text{rms}_{\epsilon_p}(\nabla A_{l,g})}{\text{rms}_{\epsilon_p}(A_{l,g})}, \quad p_{l,g}^{\text{out}} = \frac{\text{rms}_{\epsilon_p}(\nabla B_{l,g})}{\text{rms}_{\epsilon_p}(B_{l,g})}. \quad (9)$$

The coefficient-gradient energy scale for the global response is

$$p_{l,g}^R = \left[\frac{1}{2} (\text{mean}((\nabla a_{l,g,\cdot})^2) + \text{mean}((\nabla b_{l,g,\cdot})^2)) + \epsilon_p \right]^{1/2}. \quad (10)$$

The same numerical floor is used in $p_{l,g}^R$, although this quantity is raw coefficient-gradient energy rather than a relative matrix pressure. MatrixPolicy maintains positive exponential moving averages $q_{l,g}^{\text{in}}$, $q_{l,g}^{\text{out}}$, and $q_{l,g}^R$ of these quantities with decay 0.95, initialized from the first observed post-backpropagation values, and forms

$$\pi_{l,g} = \log q_{l,g}^{\text{in}} - \log q_{l,g}^{\text{out}}, \quad \alpha_{l,g} = \log q_{l,g}^R - \frac{1}{2} (\log q_{l,g}^{\text{in}} + \log q_{l,g}^{\text{out}}). \quad (11)$$

Each matrix pressure is an unsigned floored relative RMS-gradient proxy, so $\pi_{l,g}$ is signed only because it compares the input and output proxy EMAs. MatrixPolicy uses $\pi_{l,g}$ as a bounded redistribution and pair-balance signal. The score $\alpha_{l,g}$ is a fixed-scale implementation contrast between raw coefficient-gradient activity and adjacent matrix-pressure proxies; it is used only inside bounded redistribution and gating rules, not as a scale-invariant physical ratio.

The activation also provides batch-estimated curve gains. From a recent sample of normalized coordinates $u_{l,g}$ cached by the RLB forward path, MatrixPolicy uses floored summaries

$$\hat{d}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R'_{l,g}(u_{l,g})^2) + \epsilon_g}, \quad \hat{o}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R_{l,g}(u_{l,g})^2) + \epsilon_g}. \quad (12)$$

The mean is over cached scalar normalized coordinates for group (l, g) across batch, sequence positions, and the m group coordinates. The derivative gain is the input-role proxy; the response gain is

the recombination-role proxy. The latter excludes the group radius $\rho_{l,g}(x_l)$, so neither gain is a full feature scale or block-Jacobian preconditioner.

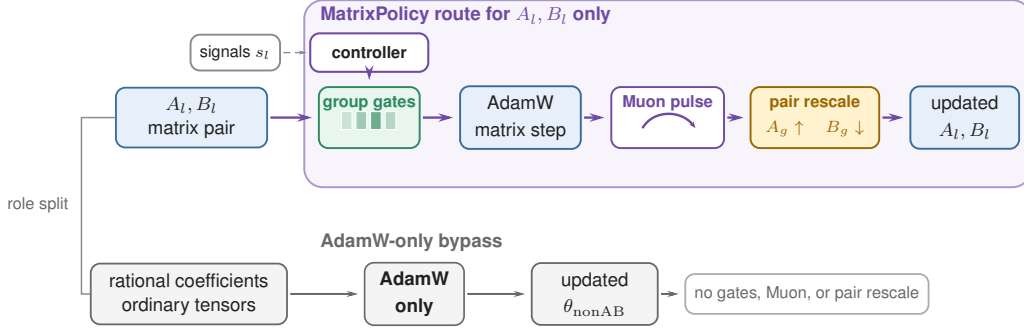


Figure 2: Selective MatrixPolicy route for the evaluated optimizer. Only the RLB matrix pair A_l, B_l enters this route: policy signals set group gradient gates $c_{l,g,\sigma}$, AdamW supplies the base matrix step, a transient Muon correction acts on the same matrices, and pair rescaling applies reciprocal updates $A_{l,g} \leftarrow e^{\ell} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell} B_{l,g}$ to move the matrix-RMS log ratio toward its target. Rational coefficients and all non-RLB matrix parameters bypass this route and remain AdamW-only.

Let t be the optimizer step, T the planned number of training steps, and $\xi_t = t/T$. A smooth gate

$$\phi_t = \text{smoothstep}(0.02, 0.30, \xi_t) \quad (13)$$

turns on the group-gradient policy. For positive group values v_j , define $\text{geomean}_{j=1}^G(v_j) = \exp\left(G^{-1} \sum_{j=1}^G \log v_j\right)$. The multiplier for group g and role $\sigma \in \{\text{in}, \text{out}\}$ is

$$c_{l,g,\sigma} = c_{l,g,\sigma}^{\text{gain}} c_{l,g,\sigma}^{\text{pressure}} c_{l,g}^{\text{activity}}. \quad (14)$$

For the input role, the gain term uses $\hat{d}_{l,g}^{\text{live}}$; for the output role, it uses $\hat{o}_{l,g}^{\text{live}}$:

$$c_{l,g,\text{in}}^{\text{gain}} = \left(\frac{\text{geomean}_{j=1}^G \hat{d}_{l,j}^{\text{live}}}{\hat{d}_{l,g}^{\text{live}}} \right)^{0.20\phi_t}, \quad c_{l,g,\text{out}}^{\text{gain}} = \left(\frac{\text{geomean}_{j=1}^G \hat{o}_{l,j}^{\text{live}}}{\hat{o}_{l,g}^{\text{live}}} \right)^{0.20\phi_t}. \quad (15)$$

With $\tilde{\pi}_{l,g} = \text{clip}(\pi_{l,g}, -1.50, 1.50)$, the relative-gradient term sets role-dependent group multipliers before the later role-wise recentering:

$$c_{l,g,\text{in}}^{\text{pressure}} = \exp(-0.10\phi_t \tilde{\pi}_{l,g}), \quad c_{l,g,\text{out}}^{\text{pressure}} = \exp(+0.10\phi_t \tilde{\pi}_{l,g}). \quad (16)$$

Coefficient-gradient activity downweights groups whose rational coefficients are moving more strongly than the adjacent matrices before role-wise recentering:

$$c_{l,g}^{\text{activity}} = \exp \left[-0.20\phi_t \text{ReLU} \left(\frac{\alpha_{l,g} - 0.05}{0.45} \right) \right]. \quad (17)$$

The final multiplier is recentered and clipped,

$$c_{l,g,\sigma} \leftarrow \frac{c_{l,g,\sigma}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}}, \quad c_{l,g,\sigma} \leftarrow \text{clip}(c_{l,g,\sigma}, 0.75, 1.35), \quad (18)$$

so the policy redistributes update scale across groups, separately for each matrix role, without changing the role-wise mean log multiplier before clipping.

The scaled matrix gradients are

$$\tilde{\nabla} A_{l,g} = c_{l,g,\text{in}} \nabla A_{l,g}, \quad \tilde{\nabla} B_{l,g} = c_{l,g,\text{out}} \nabla B_{l,g}. \quad (19)$$

Let $\tilde{\nabla} M_{l,\text{in}}$ be the row-block concatenation of $\{\tilde{\nabla} A_{l,g}\}_{g=1}^G$, and let $\tilde{\nabla} M_{l,\text{out}}$ be the column-block concatenation of $\{\tilde{\nabla} B_{l,g}\}_{g=1}^G$. After this scaling, the RLB matrices are updated by a role/depth

AdamW branch and a transient Muon branch. Define $\delta_l = (l - 1)/(L - 1)$ for $L > 1$ and $\delta_l = 1/2$ otherwise. The role factors are

$$\varrho_{\text{in}}(l) = \text{clip}(1 - 0.50(\delta_l - 0.5), 0.55, 1.40), \quad \varrho_{\text{out}}(l) = \text{clip}(1 + 1.00(\delta_l - 0.5), 0.55, 1.40), \quad (20)$$

and the AdamW matrix multiplier for role σ is

$$s_{l,\sigma}^{\text{Adam}} = \text{clip}(3.0 \max\{0.10, 1 + 1.20(\varrho_{\sigma}(l) - 1)\}, 0.40, 4.0). \quad (21)$$

The Muon branch is concentrated early in training and is reduced when relative scale imbalance or coefficient-gradient scale is high. Define

$$\Psi_{l,t} = \text{clip}\left(\frac{1}{G} \sum_{g=1}^G |\pi_{l,g}|, 0, 1.50\right), \quad \Omega_{l,t} = \text{ReLU}\left(\frac{G^{-1} \sum_{g=1}^G \alpha_{l,g} - 0.05}{0.45}\right), \quad (22)$$

$$\psi_{l,t} = \text{clip}(\exp[-0.30\Psi_{l,t} - 0.65\Omega_{l,t}], 0.10, 1.15). \quad (23)$$

The branch gate is

$$\mu_{l,\sigma,t} = \text{clip}(0.75 \text{ smoothstep}(0.02, 0.12, \xi_t)[1 - \text{smoothstep}(0.20, 0.36, \xi_t)]\varrho_{\sigma}(l)\psi_{l,t}, 0, 0.75). \quad (24)$$

For $M_{l,\text{in}} = A_l$ and $M_{l,\text{out}} = B_l$, let η_t be the base learning rate. MatrixPolicy applies an AdamW matrix step followed by any active Muon substep. The gate $\mu_{l,\sigma,t}$ scales the two learning rates; it is not a convex mixing coefficient for a single update direction:

$$\begin{aligned} \widetilde{M}_{l,\sigma}^{t+1} &= \text{AdamW}_{\eta_t s_{l,\sigma}^{\text{Adam}}(1-\mu_{l,\sigma,t})}(M_{l,\sigma}^t; \widetilde{\nabla} M_{l,\sigma}^t), \\ M_{l,\sigma}^{t+1} &= \text{Muon}_{\eta_t \mu_{l,\sigma,t}}(\widetilde{M}_{l,\sigma}^{t+1}; \widetilde{\nabla} M_{l,\sigma}^t), \end{aligned} \quad (25)$$

The update in equation 25 is operational notation: AdamW and Muon carry separate optimizer states, their configured weight-decay and momentum terms are handled by the respective optimizer implementations, and both substeps use the gradient from training step t . Here Muon denotes the `torch.optim.Muon` update on two-dimensional RLB matrix parameters: momentum 0.95, five Newton–Schulz orthogonalization steps, and the match-RMS learning-rate adjustment of the Muon optimizer (Liu et al., 2025). When all Muon fractions have decayed to zero, this branch is inactive.

The last operation is a bounded positive rescaling of the two matrix blocks around each rational curve. It is evaluated after the AdamW/Muon matrix branch every five optimizer steps. Let $\mathcal{U} = \{u_k\}_{k=1}^{129}$ be a uniform probe grid over $[-5, 5]$, and define $\text{rms}_{\mathcal{U}}(\varphi) = \sqrt{|\mathcal{U}|^{-1} \sum_{u_k \in \mathcal{U}} \varphi(u_k)^2}$. Fixed-grid probe gains describe curve shape for pair balancing; live gains in equation 12 describe the current batch distribution for group-gradient redistribution. The probe gains are

$$\hat{d}_{l,g}^{\text{probe}} = \sqrt{|\mathcal{U}|^{-1} \sum_{u_k \in \mathcal{U}} R'_{l,g}(u_k)^2 + \epsilon_g}, \quad \hat{o}_{l,g}^{\text{probe}} = \sqrt{|\mathcal{U}|^{-1} \sum_{u_k \in \mathcal{U}} R_{l,g}(u_k)^2 + \epsilon_g}, \quad (26)$$

and the current matrix log ratio is computed from the post-branch matrix blocks with floored matrix RMS values,

$$\gamma_{l,g} = \log \text{rms}_{\epsilon_p}(A_{l,g}) - \log \text{rms}_{\epsilon_p}(B_{l,g}). \quad (27)$$

Because the floor is inside both logarithms, $\gamma_{l,g}$ is a stable implementation proxy for a matrix-RMS ratio. The target ratio combines curve-gain and relative-gradient terms,

$$\tau_{l,g} = \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} \right) + 0.25 \bar{\pi}_{l,g}, \quad \bar{\pi}_{l,g} = \text{clip}(\pi_{l,g}, -1.25, 1.25). \quad (28)$$

The first term balances fixed-grid derivative and response gains in log space, with the factor $1/2$ mapping the gain contrast to the matrix-pair log-ratio coordinate; the second term adds a smaller pressure bias from the clipped RMS-imbalance proxy. Coefficient-gradient scale modulates the rescaling strength through

$$\chi_{l,g}^{\text{rs}} = \exp(0.10 \bar{\alpha}_{l,g}), \quad \bar{\alpha}_{l,g} = \text{clip}(\alpha_{l,g}, -1.25, 1.25). \quad (29)$$

High group coefficient activity damps the group-gradient path, while high layer-average coefficient activity damps the Muon branch; in the bounded rescaling step it instead moves the pair slightly faster toward the representative ratio, with the log-step clip limiting the effect. With schedule

$$s_{l,t} = 0.50 \text{ smoothstep}(0.03, 0.35, \xi_t) \text{ clip}(1 + 0.15(\delta_l - 0.5), 0.75, 1.25), \quad (30)$$

the active-step log move is

$$\ell_{l,g,t} = \text{clip}\left(\frac{1}{2}[\pi_{l,g} - \gamma_{l,g}]s_{l,t}\chi_{l,g}^{\text{rs}}, -0.030, 0.030\right). \quad (31)$$

The matrix blocks are then updated as

$$A_{l,g} \leftarrow e^{\ell_{l,g,t}} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_{l,g,t}} B_{l,g}. \quad (32)$$

Because equation 27 uses floored matrix RMS values, this update is an exact log-ratio move only in the nonzero, unfloored idealization; near the floor it is a clipped balance step. The implementation applies the corresponding group scale to optimizer-state buffers as well, using squared scales for square-average state entries.

4 EXPERIMENTS

4.1 FIXED-SCALE TARGET-ARRIVAL STUDY

The fixed-scale study uses the same 12-layer, width-768 causal Transformer on five corpora: DCLM/DataComp-LM (Li et al., 2024), FineWeb-Edu and FineWeb (Penedo et al., 2024), Dolma-sample (Soldaini et al., 2024), and C4 (Raffel et al., 2020). The 100M-token budget uses 3050 optimizer steps and the 300M-token budget uses 9150 optimizer steps, with three seeds per method/dataset pair. The main comparison keeps the architecture and data fixed while changing the activation and optimizer used in the nonlinear sublayer: MatrixPolicy is compared with SiLU controls and with RLB-only controls, where RLB-only uses the same global-rational RLB activation but no MatrixPolicy updates. Shared optimizer, evaluation-cadence, and throughput-measurement details are reported in Appendix A.1; broader optimizer comparisons beyond AdamW and Muon are summarized in Table 2.

Table 1: Target-arrival efficiency for the fixed 12-layer, width-768 Transformer. RLB denotes the global-rational RLB activation without MatrixPolicy. At the hardest validation-loss target reached by every listed method in all three seeds, MatrixPolicy is always fastest; each token-saving column reports MatrixPolicy’s savings relative to the method named below it.

Budget	Dataset	Target loss	MatrixPolicy token savings (%)				Time to target (min)				
			vs. SiLU AdamW	vs. SiLU Muon	vs. RLB AdamW	vs. RLB Muon	Matrix Policy	SiLU AdamW	SiLU Muon	RLB AdamW	RLB Muon
100M	DCLM	4.55	20.7	32.4	20.7	34.3	12.8	20.8	26.0	14.3	19.0
	FineWeb-Edu	4.40	19.5	29.5	20.2	29.5	12.9	20.3	24.4	14.2	18.0
	FineWeb	4.60	22.8	32.1	21.5	32.1	12.6	16.4	20.2	16.2	20.1
	Dolma	4.60	22.0	34.0	22.7	34.9	12.8	17.6	22.0	17.3	22.2
	C4	4.60	20.7	29.2	19.3	28.7	11.6	14.3	17.3	14.9	18.8
300M	DCLM	4.10	23.3	8.3	24.0	7.0	39.6	53.5	45.9	54.7	45.6
	FineWeb-Edu	3.85	22.8	6.7	22.6	4.8	40.0	59.9	47.9	52.3	50.0
	FineWeb	4.10	23.1	7.1	22.4	5.3	40.8	61.2	46.4	55.6	46.6
	Dolma	3.95	22.3	9.8	22.3	6.5	44.4	49.3	52.3	50.5	47.4
	C4	4.00	26.4	8.5	25.3	6.5	41.8	64.7	55.5	59.4	51.9

Table 1 is the primary empirical result. It asks: for the hardest common validation-loss target reached by every listed method in all three seeds, what fraction of target-arrival tokens does MatrixPolicy save, and how many measured target-arrival minutes does each method require? Across all five datasets under the 100M-token budget, MatrixPolicy saves 19.5–22.8% of the SiLU with AdamW

target-arrival tokens and 29.2–34.0% of the SiLU with Muon target-arrival tokens. Under the 300M-token budget, after longer training compresses endpoint gaps, MatrixPolicy still saves 22.3–26.4% versus SiLU with AdamW and 6.7–9.8% versus SiLU with Muon, with lower target-arrival time in every listed comparison.

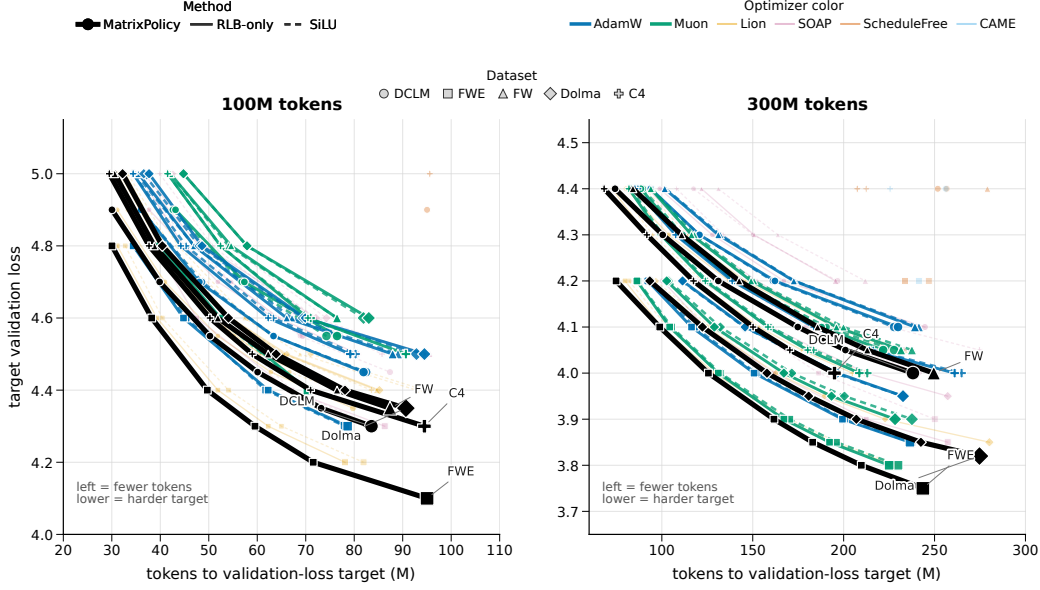


Figure 3: Target-arrival map for the completed fixed-scale study. Each point is the mean token count needed by one method to reach one validation-loss target on one dataset; connected points trace the same method and dataset from easier to harder targets. The horizontal coordinate is target-arrival tokens, and the vertical coordinate is the target validation loss, so curves further left reach the same loss with fewer tokens. The hard-target timing and token-saving percentages are quantified in Table 1.

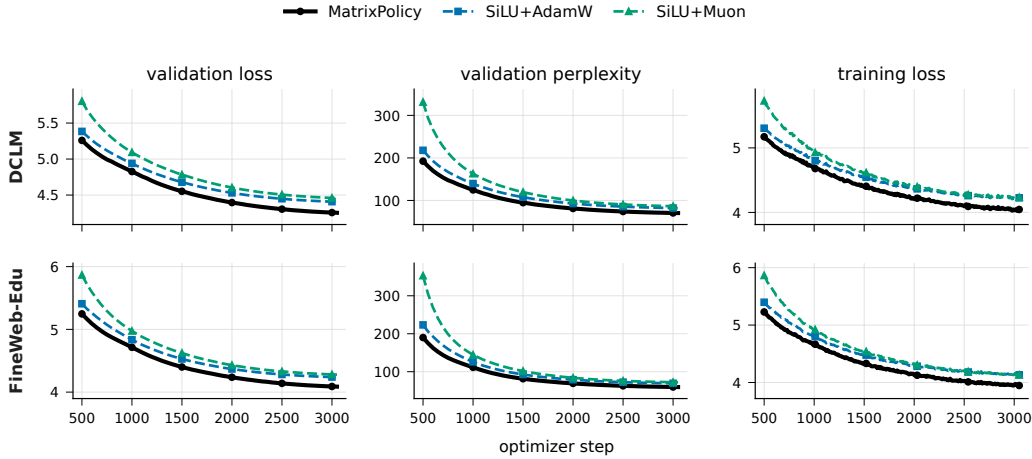


Figure 4: Representative 100M-token-budget training dynamics on DCLM and FineWeb-Edu. Rows show datasets and columns show validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. This figure uses the two standard SiLU baselines so that the early target-arrival effect in Table 1 remains visually legible.

The 100M-token-budget curves show why target arrival is the right main readout. MatrixPolicy separates early in validation loss and validation perplexity while keeping training loss competitive. A long fixed budget eventually compresses many activation/optimizer endpoints, especially under the 300M-token budget, so the paper emphasizes the number of tokens and minutes needed to reach the same useful validation-loss levels rather than only the final loss after every method has consumed the full budget.

4.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert the larger-scale training result here.

4.3 TODO: COMPONENT ABLATIONS

TODO: Insert the component ablation study here.

5 CONCLUSION

Global-rational RLB turns the nonlinear Transformer sublayer into an optimizer-visible interface: it exposes grouped normalized coordinates, trainable global rational responses, and an idealized positive matrix-pair relation used as a bounded balancing coordinate. MatrixPolicy uses that interface to update the RLB input and output matrices with role-aware statistics, while rational coefficients and non-RLB-matrix parameters use AdamW. The completed 100M-token and 300M-token runs show a consistent target-arrival pattern: MatrixPolicy reaches common validation losses with fewer tokens and lower measured target-arrival time than the listed SiLU and global-rational controls. [todo: Complete this conclusion after the larger-scale and ablation results are added.]

REFERENCES

- Nicolas Boullé, Yuji Nakatsukasa, and Alex Townsend. Rational Neural Networks. In *Advances in Neural Information Processing Systems*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/a3f390d88e4c41f2747bfa2f1b5f87db-Abstract.html>.
- Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Pham, Xuanyi Dong, Thang Lu, Cho-Jui Hsieh, Yifeng Lu, and Quoc V. Le. Symbolic Discovery of Optimization Algorithms. In *Advances in Neural Information Processing Systems*, 2023. URL https://papers.neurips.cc/paper_files/paper/2023/hash/9a39b4925e35cf447ccba8757137d84f-Abstract-Conference.html.
- Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, et al. PaLM: Scaling Language Modeling with Pathways. *arXiv preprint arXiv:2204.02311*, 2022. URL <https://arxiv.org/abs/2204.02311>.
- Yann N. Dauphin, Angela Fan, Michael Auli, and David Grangier. Language Modeling with Gated Convolutional Networks. In *Proceedings of the 34th International Conference on Machine Learning*, pp. 933–941, 2017. URL <https://proceedings.mlr.press/v70/dauphin17a.html>.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chengqi Deng, Chong Ruan, Damai Dai, et al. DeepSeek-V3 Technical Report. *arXiv preprint arXiv:2412.19437*, 2024. URL <https://arxiv.org/abs/2412.19437>.
- Aaron Defazio, Xingyu Yang, Ahmed Khaled, Konstantin Mishchenko, Harsh Mehta, and Ashok Cutkosky. The Road Less Scheduled. In *Advances in Neural Information Processing Systems*, 2024. URL <https://neurips.cc/virtual/2024/poster/96925>.
- Haishuo Fang, Ji-Ung Lee, Nafise Sadat Moosavi, and Iryna Gurevych. Transformers with Learnable Activation Functions. In *Findings of the Association for Computational Linguistics: EACL 2023*, 2023. URL <https://aclanthology.org/2023.findings-eacl.181/>.

- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned Stochastic Tensor Optimization. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 1842–1850, 2018. URL <https://proceedings.mlr.press/v80/gupta18a.html>.
- Dan Hendrycks and Kevin Gimpel. Gaussian Error Linear Units (GELUs). *arXiv preprint arXiv:1606.08415*, 2016. URL <https://arxiv.org/abs/1606.08415>.
- Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*, 2015. URL <https://mlanthology.org/iclr/2015/kingma2015iclr-adam/>.
- Houyi Li, Wenzhen Zheng, Qiufeng Wang, Zhenyu Ding, Haoying Wang, Zili Wang, Shijie Xuyang, Ning Ding, Shuigeng Zhou, Xiangyu Zhang, and Daxin Jiang. Predictable Scale: Part II, Farseeer – A Refined Scaling Law in Large Language Models. *arXiv preprint arXiv:2506.10972*, 2025. URL <https://arxiv.org/abs/2506.10972>.
- Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, Hritik Bansal, Etash Guha, et al. DataComp-LM: In Search of the Next Generation of Training Sets for Language Models. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-0455. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/19e4ea30dded58259665db375885e412-Abstract-Datasets_and_Benchmarks_Track.html.
- Kaizhao Liang, Lizhang Chen, Bo Liu, and Qiang Liu. Cautious Optimizers: Improving Training with One Line of Code. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=zBPZeRjfgu>.
- Fengze Liu, Weidong Zhou, Binbin Liu, Ping Guo, Zijun Wang, Bingni Zhang, Yifan Zhang, Yifeng Yu, Xiaohuan Zhou, and Taifeng Wang. InfoLaw: Information Scaling Laws for Large Language Models with Quality-Weighted Mixture Data and Repetition. *arXiv preprint arXiv:2605.02364*, 2026. URL <https://arxiv.org/abs/2605.02364>.
- Hong Liu, Zhiyuan Li, David Hall, Percy Liang, and Tengyu Ma. Sophia: A Scalable Stochastic Second-order Optimizer for Language Model Pre-training. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/06960915ba8674c7a898ec0b472b80ff-Abstract-Conference.html.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is Scalable for LLM Training. *arXiv preprint arXiv:2502.16982*, 2025. URL <https://arxiv.org/abs/2502.16982>.
- Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Yang Luo, Xiaozhe Ren, Zangwei Zheng, Zhuo Jiang, Xin Jiang, and Yang You. CAME: Confidence-guided Adaptive Memory Efficient Optimization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pp. 4442–4453, 2023. doi: 10.18653/v1/2023.acl-long.243. URL <https://aclanthology.org/2023.acl-long.243/>.
- Alejandro Molina, Patrick Schramowski, and Kristian Kersting. Padé Activation Units: End-to-End Learning of Flexible Activation Functions in Deep Networks. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=BJlBSkHtDS>.

- Guilherme Penedo, Hynek Kydlicek, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-0970. URL https://papers.nips.cc/paper/2024/hash/370df50ccfdf8bde18f8f9c2d9151bda-Abstract-Datasets_and_Benchmarks_Track.html.
- Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115*, 2025. URL <https://arxiv.org/abs/2412.15115>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <http://jmlr.org/papers/v21/20-074.html>.
- Prajit Ramachandran, Barret Zoph, and Quoc V. Le. Searching for Activation Functions. *arXiv preprint arXiv:1710.05941*, 2017. URL <https://arxiv.org/abs/1710.05941>.
- Andrei Semenov, Matteo Pagliardini, and Martin Jaggi. Benchmarking Optimizers for Large Language Model Pretraining. *arXiv preprint arXiv:2509.01440*, 2025. URL <https://arxiv.org/abs/2509.01440>.
- Noam Shazeer. GLU Variants Improve Transformer. *arXiv preprint arXiv:2002.05202*, 2020. URL <https://arxiv.org/abs/2002.05202>.
- Noam Shazeer and Mitchell Stern. Adafactor: Adaptive Learning Rates with Sublinear Memory Cost. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 4596–4604, 2018. URL <https://proceedings.mlr.press/v80/shazeer18a.html>.
- Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, et al. Dolma: An Open Corpus of Three Trillion Tokens for Language Model Pretraining Research. *arXiv preprint arXiv:2402.00159*, 2024. URL <https://arxiv.org/abs/2402.00159>.
- Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv preprint arXiv:2104.09864*, 2021. URL <https://arxiv.org/abs/2104.09864>.
- Maosen Tang and Alex Townsend. Rational Neural Networks have Expressivity Advantages. *arXiv preprint arXiv:2602.12390*, 2026. doi: 10.48550/arXiv.2602.12390. URL <https://arxiv.org/abs/2602.12390>.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, et al. LLaMA: Open and Efficient Foundation Language Models. *arXiv preprint arXiv:2302.13971*, 2023. URL <https://arxiv.org/abs/2302.13971>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 2017. URL https://papers.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.
- Alexandra Volkova, Mher Safaryan, Christoph H. Lampert, and Dan Alistarh. Towards Robust Scaling Laws for Optimizers. *arXiv preprint arXiv:2602.07712*, 2026. URL <https://arxiv.org/abs/2602.07712>.
- Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham Kakade. SOAP: Improving and Stabilizing Shampoo using Adam for Language Modeling. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e988664070e9591f93fdcf605f7dc623-Abstract-Conference.html.

Kaiyue Wen, David Leo Wright Hall, Tengyu Ma, and Percy Liang. Fantastic Pretraining Optimizers and Where to Find Them. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=2J51qUZ0iG>.

Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. *arXiv preprint arXiv:1910.07467*, 2019. URL <https://arxiv.org/abs/1910.07467>.

Yushun Zhang, Congliang Chen, Ziniu Li, Tian Ding, Chenwei Wu, Diederik P. Kingma, Yinyu Ye, Zhi-Quan Luo, and Ruoyu Sun. Adam-mini: Use Fewer Learning Rates To Gain More. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/45ae878717399e6f62d57c65f052cd46-Abstract-Conference.html.

Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuandong Tian. GaLore: Memory-Efficient LLM Training by Gradient Low-Rank Projection. In *International Conference on Machine Learning*, 2024. URL <https://openreview.net/forum?id=hYHsrKDIX7>.

Rosie Zhao, Depen Morwani, David Brandfonbrener, Nikhil Vyas, and Sham Kakade. Deconstructing What Makes a Good Optimizer for Language Models. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2407.07972>.

A EXPERIMENT APPENDIX

A.1 FIXED-SCALE TARGET-ARRIVAL STUDY

A.1.1 MODEL AND DATA PIPELINE

The fixed-scale experiments use a custom causal decoder Transformer rather than an off-the-shelf GPT-2 or LLaMA implementation. The block uses a LLaMA-style pre-normalized layout with RMSNorm before attention and before the nonlinear sublayer, bias-free query/key/value and output attention projections, rotary position embeddings, causal scaled-dot-product attention, residual attention and nonlinear-sublayer updates, a final RMSNorm, and a tied token-embedding/output head (Zhang & Sennrich, 2019; Su et al., 2021; Touvron et al., 2023). The fixed model has 12 layers, width 768, 12 heads of dimension 64, and context length 256.

The SiLU controls use a SwiGLU-like gated nonlinear sublayer with bias-free matrices $W_g, W_v \in \mathbb{R}^{768 \times 2048}$ and $W_o \in \mathbb{R}^{2048 \times 768}$, computing $(\text{SiLU}(xW_g) \odot xW_v)W_o$. The global-rational RLB rows use a matched single-branch matrix pair: an input matrix maps width 768 to 3072 latent coordinates, the coordinates are partitioned into groups and transformed by the real-pole-free P5/Q4 global rational response, and an output matrix maps the 3072 coordinates back to width 768. With the default group size 256, this gives $G = 12$ groups of width $m = 256$; the RMS floor is $\epsilon_{\text{rms}} = 10^{-6}$. This is the global-rational RLB variant used by the main MatrixPolicy results.

Tokenization uses the GPT-2 tokenizer through the Hugging Face tokenizer interface, without adding tokenizer-specific special tokens; an EOS or pad token is appended between documents. Training samples random contiguous blocks of length 257 and forms next-token prediction pairs of length 256. The DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 runs use the text field specified in the manifests, with validation slices drawn from held-out token offsets of the same corpus stream.

A.1.2 TRAINING PROTOCOL

All runs use batch size 16, gradient accumulation 2, and therefore 32768 global tokens per optimizer step. The 100M-token budget consists of 3050 optimizer steps; the 300M-token budget consists of 9150 optimizer steps. Each method/dataset pair is run with seeds 1337, 2027, and 3407. Validation is performed every 50 optimizer steps, so token-to-target arrivals are quantized in increments of 1.64M tokens.

The MatrixPolicy rows in the main table use the global-rational RLB activation: the active nonlinearity is the group-specific global P5/Q4 response in equations 3–4. The evaluated configuration uses AdamW as the backbone optimizer and disables the separate rational-coefficient optimizer.

It enables MatrixPolicy group redistribution with gain/pressure/activity strengths 0.20/0.10/0.20, schedule window 0.02–0.30, and group multiplier clip $[0.75, 1.35]$. The bounded RLB pair-rescale wrapper runs every five optimizer steps with covariant optimizer state scaling. Rational coefficients, attention weights, embeddings, normalization parameters, and ordinary Transformer weights use AdamW in this configuration.

Target-arrival minutes in Table 1 are computed seed-wise from the first validation event that reaches the shared target and the same run’s measured seconds per optimizer step.

A.1.3 SUPPORTING FIXED-SCALE CURVES

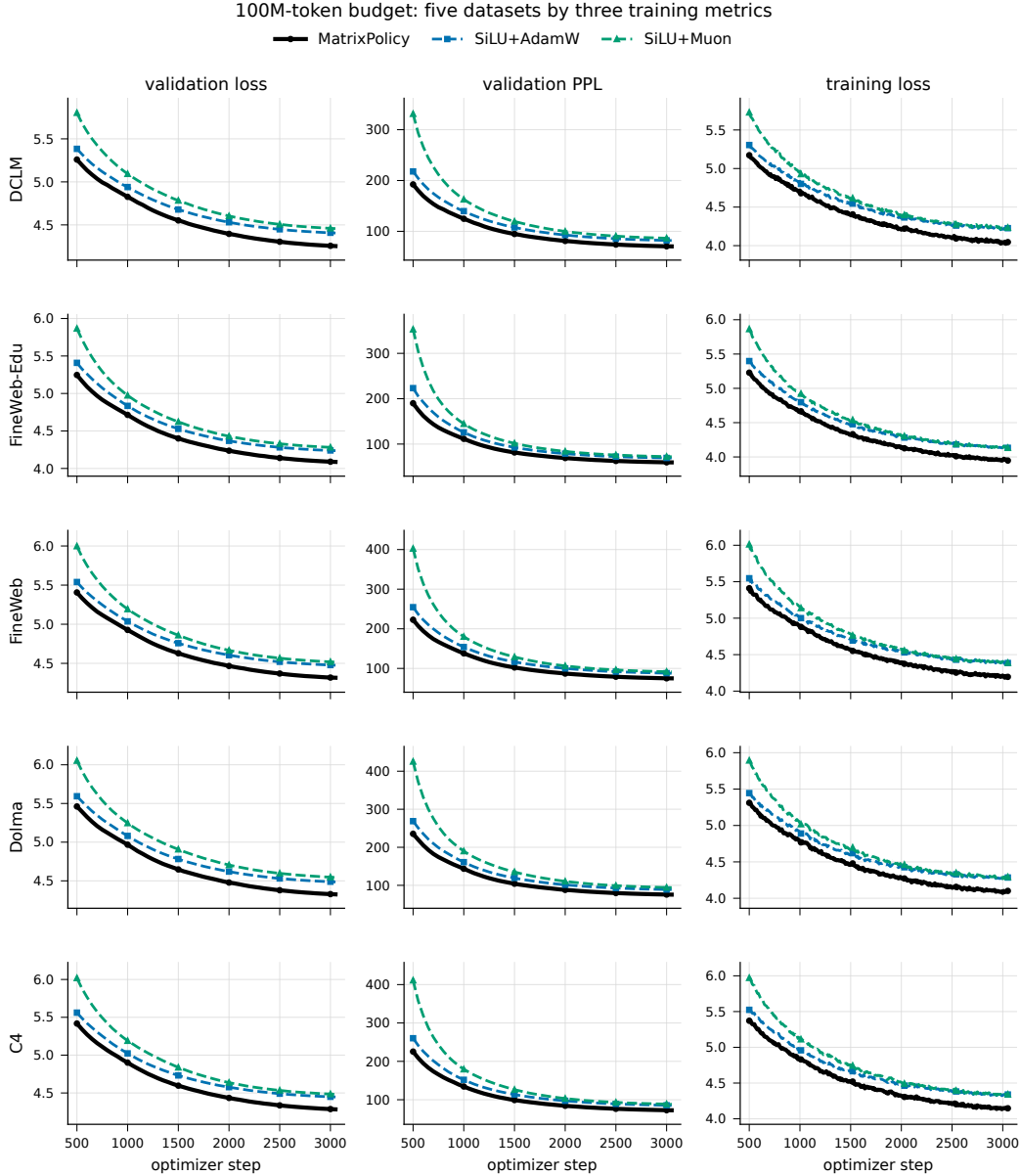


Figure 5: 100M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation.

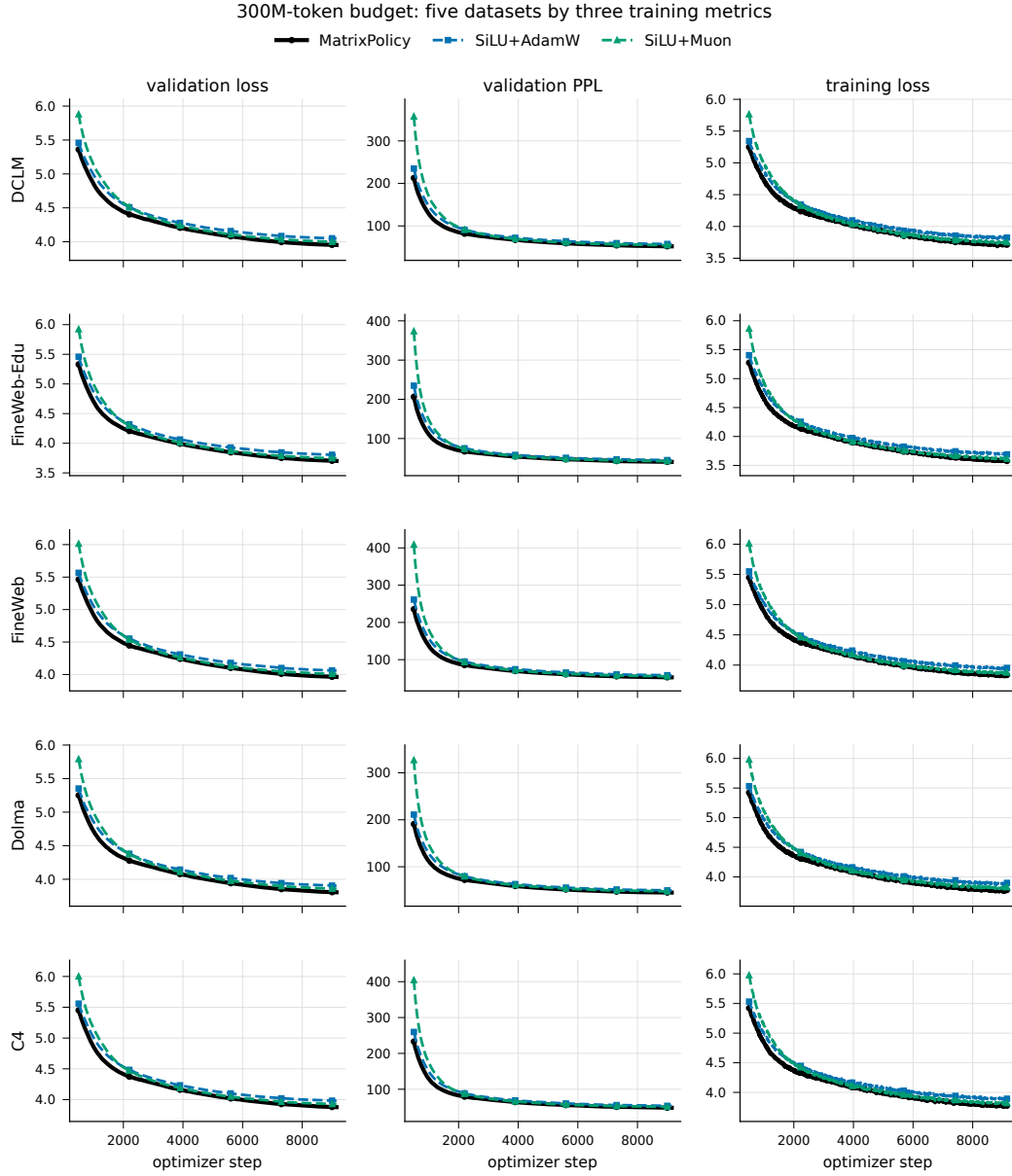


Figure 6: 300M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation.

Table 2: Final validation loss for the broader optimizer sweep on the fixed 12-layer, width-768 language model. Values are mean $\pm$ sample standard deviation over three seeds; lower is better. RLB rows use the global-rational RLB activation without MatrixPolicy.

Budget	Method	DCLM	FineWeb-Edu	FineWeb	Dolma	C4
100M tokens	MatrixPolicy	4.2538 $\pm$ 0.0063	4.0873 $\pm$ 0.0102	4.3162 $\pm$ 0.0125	4.3253 $\pm$ 0.0053	4.2837 $\pm$ 0.0197
	SiLU + AdamW	4.4056 $\pm$ 0.0099	4.2375 $\pm$ 0.0086	4.4758 $\pm$ 0.0097	4.4862 $\pm$ 0.0012	4.4469 $\pm$ 0.0160
	SiLU + Lion	4.3183 $\pm$ 0.0069	4.1494 $\pm$ 0.0092	4.3825 $\pm$ 0.0083	4.3878 $\pm$ 0.0046	4.3536 $\pm$ 0.0156
	SiLU + Muon	4.4572 $\pm$ 0.0126	4.2787 $\pm$ 0.0243	4.5163 $\pm$ 0.0264	4.5443 $\pm$ 0.0108	4.4824 $\pm$ 0.0233
	SiLU + SOAP	4.4160 $\pm$ 0.0038	4.2630 $\pm$ 0.0220	4.4840 $\pm$ 0.0112	4.4987 $\pm$ 0.0035	4.4582 $\pm$ 0.0146
	SiLU + ScheduleFree	4.9023 $\pm$ 0.0111	4.8258 $\pm$ 0.0072	5.0142 $\pm$ 0.0188	5.0494 $\pm$ 0.0149	5.0064 $\pm$ 0.0160
	SiLU + CAME	5.0107 $\pm$ 0.0143	4.9207 $\pm$ 0.0123	5.1325 $\pm$ 0.0126	5.1650 $\pm$ 0.0189	5.1230 $\pm$ 0.0178
	RLB + AdamW	4.4046 $\pm$ 0.0081	4.2392 $\pm$ 0.0077	4.4717 $\pm$ 0.0138	4.4878 $\pm$ 0.0027	4.4412 $\pm$ 0.0162
	RLB + Lion	4.2946 $\pm$ 0.0083	4.1361 $\pm$ 0.0083	4.3626 $\pm$ 0.0112	4.3622 $\pm$ 0.0066	4.3271 $\pm$ 0.0160
	RLB + Muon	4.4674 $\pm$ 0.0121	4.2831 $\pm$ 0.0284	4.5157 $\pm$ 0.0059	4.5476 $\pm$ 0.0153	4.4764 $\pm$ 0.0099
	RLB + SOAP	4.4360 $\pm$ 0.0442	4.2747 $\pm$ 0.0238	4.6893 $\pm$ 0.3359	4.5183 $\pm$ 0.0363	4.4458 $\pm$ 0.0190
	RLB + ScheduleFree	4.8871 $\pm$ 0.0046	4.7957 $\pm$ 0.0133	4.9993 $\pm$ 0.0192	5.0363 $\pm$ 0.0129	4.9842 $\pm$ 0.0164
	RLB + CAME	5.0139 $\pm$ 0.0086	4.9150 $\pm$ 0.0063	5.1340 $\pm$ 0.0162	5.1514 $\pm$ 0.0168	5.1109 $\pm$ 0.0155
300M tokens	MatrixPolicy	3.9518 $\pm$ 0.0282	3.7015 $\pm$ 0.0212	3.9623 $\pm$ 0.0081	3.8062 $\pm$ 0.0073	3.8777 $\pm$ 0.0144
	SiLU + AdamW	4.0493 $\pm$ 0.0275	3.8035 $\pm$ 0.0182	4.0612 $\pm$ 0.0101	3.9037 $\pm$ 0.0091	3.9811 $\pm$ 0.0128
	SiLU + Lion	3.9934 $\pm$ 0.0230	3.7440 $\pm$ 0.0208	4.0015 $\pm$ 0.0085	3.8475 $\pm$ 0.0094	3.9213 $\pm$ 0.0105
	SiLU + Muon	3.9973 $\pm$ 0.0305	3.7454 $\pm$ 0.0170	4.0066 $\pm$ 0.0128	3.8581 $\pm$ 0.0101	3.9251 $\pm$ 0.0134
	SiLU + SOAP	4.0964 $\pm$ 0.0300	3.8629 $\pm$ 0.0201	4.1139 $\pm$ 0.0101	3.9568 $\pm$ 0.0093	4.0349 $\pm$ 0.0108
	SiLU + ScheduleFree	4.3657 $\pm$ 0.0298	4.1559 $\pm$ 0.0238	4.3979 $\pm$ 0.0106	4.2151 $\pm$ 0.0053	4.3163 $\pm$ 0.0107
	SiLU + CAME	4.3682 $\pm$ 0.0226	4.1503 $\pm$ 0.0211	4.4060 $\pm$ 0.0189	4.2492 $\pm$ 0.0270	4.3298 $\pm$ 0.0148
	RLB + AdamW	4.0496 $\pm$ 0.0298	3.8024 $\pm$ 0.0206	4.0601 $\pm$ 0.0110	3.9045 $\pm$ 0.0082	3.9787 $\pm$ 0.0098
	RLB + Lion	3.9887 $\pm$ 0.0295	3.7416 $\pm$ 0.0214	3.9960 $\pm$ 0.0105	3.8412 $\pm$ 0.0085	3.9132 $\pm$ 0.0139
	RLB + Muon	3.9913 $\pm$ 0.0260	3.7373 $\pm$ 0.0187	3.9991 $\pm$ 0.0110	3.8478 $\pm$ 0.0047	3.9189 $\pm$ 0.0149
	RLB + SOAP	4.0608 $\pm$ 0.0331	3.8240 $\pm$ 0.0128	4.1081 $\pm$ 0.0320	3.9269 $\pm$ 0.0125	4.0024 $\pm$ 0.0194
	RLB + ScheduleFree	4.3607 $\pm$ 0.0344	4.1421 $\pm$ 0.0217	4.3864 $\pm$ 0.0081	4.2121 $\pm$ 0.0112	4.3084 $\pm$ 0.0071
	RLB + CAME	4.4503 $\pm$ 0.0426	4.2255 $\pm$ 0.0358	4.4804 $\pm$ 0.0064	4.2665 $\pm$ 0.0409	4.3651 $\pm$ 0.0582

A.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert larger-scale training details here.

A.3 TODO: COMPONENT ABLATIONS

TODO: Insert component ablation details here.

B RATIONAL LATENT BASIS CALCULUS

This section records the identities behind the RLB quantities used by MatrixPolicy. All calculations are local to one layer and one group unless otherwise stated, so we suppress (l, g) in the notation. The response parameterization first guarantees that the denominator cannot vanish on the real line:

$$\begin{aligned} \phi_1(u) &= |u|, \quad \phi_2(u) = u^2, \quad \phi_3(u) = |u|^3, \quad \phi_4(u) = u^4, \\ P(u) &= \sum_{i=0}^5 a_i u^i, \quad Q(u) = 1 + \sum_{j=1}^4 |b_j| \phi_j(u). \end{aligned} \quad (33)$$

Since $\phi_j(u) \geq 0$ for all real u , $Q(u) \geq 1$ and $R(u) = P(u)/Q(u)$ has no real pole. The same parameterization keeps the curve derivatives visible to the optimizer. For all $u \neq 0$, the input derivative is

$$R'(u) = \frac{P'(u)Q(u) - P(u)Q'(u)}{Q(u)^2}, \quad (34)$$

where

$$P'(u) = \sum_{i=1}^5 i a_i u^{i-1}, \quad Q'(u) = |b_1| \text{sign}(u) + 2|b_2|u + 3|b_3|u|u| + 4|b_4|u^3. \quad (35)$$

Derivative-gain statistics use the autodiff convention $\text{sign}(0) = 0$ at the absolute-value kink. The coefficient features are explicit:

$$\frac{\partial R(u)}{\partial a_i} = \frac{u^i}{Q(u)}, \quad \frac{\partial R(u)}{\partial b_j} = -\frac{P(u)}{Q(u)^2} \text{sign}(b_j) \phi_j(u), \quad (36)$$

where $\text{sign}(0) = 0$ is also used at $b_j = 0$. The response factor $R(u)$, the input derivative $R'(u)$, and the coefficient features in equation 36 are all observable from the group-specific global curve. These coefficient derivatives are classical away from $b_j = 0$ and use the stated autodiff subgradient convention at $b_j = 0$.

The RMS-normalized group map turns these curve quantities into matrix-role signals. For a group vector $z \in \mathbb{R}^m$, define

$$r(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u(z) = z/r(z), \quad h(z) = r(z)R(u(z)), \quad (37)$$

where R is applied coordinatewise. The following differentials are taken on the domain $r(z) > 0$, equivalently $\epsilon_{\text{rms}} > 0$ or $\epsilon_{\text{rms}} = 0$ with $z \neq 0$. The radius derivative is

$$dr = \frac{z^\top dz}{mr} = \frac{u^\top dz}{m}, \quad (38)$$

and the normalized-coordinate derivative is

$$du = \frac{1}{r} \left(I - \frac{uu^\top}{m} \right) dz. \quad (39)$$

The factor $I - uu^\top/m$ is exact in equation 39; it becomes the orthogonal tangent projector to the RMS sphere only in the unfloored case where $\epsilon_{\text{rms}} = 0$ and $z \neq 0$. Away from absolute-value kinks of the scalar curve, the group Jacobian is

$$J_h(z) := \frac{\partial h}{\partial z} = R(u) \frac{u^\top}{m} + \text{diag}(R'(u)) \left(I - \frac{uu^\top}{m} \right). \quad (40)$$

At nonsmooth coordinates, equation 40 uses the same autodiff subgradient convention as equation 34 and equation 36. The response vector $R(u)$ controls the curve factor inside the realized feature $h = rR(u)$, while $R'(u)$ appears in the normalized-coordinate component of the input Jacobian. The MatrixPolicy summaries $\hat{o}$ and $\hat{d}$ are therefore partial role signals: $\hat{o}$ omits the group radius r , and $\hat{d}$ omits the radial term $R(u)u^\top/m$, the output block $B_{l,g}$, and the upstream gradient. These equations expose a compact set of role signals for the adjacent matrices without requiring a full block-Jacobian preconditioner.

The same normalization also exposes a positive matrix-pair coordinate. Set $\epsilon_{\text{rms}} = 0$ and take $z \neq 0$ and $\lambda > 0$. Then

$$r(\lambda z) = \lambda r(z), \quad u(\lambda z) = u(z), \quad h(\lambda z) = \lambda h(z). \quad (41)$$

Scaling $A_{l,g}$ by λ and the matching column block $B_{l,g}$ by λ^{-1} therefore leaves the group contribution unchanged. Applying this independently to all groups gives equation 7.

With $\epsilon_{\text{rms}} > 0$, define the floored maps

$$r_\epsilon(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u_\epsilon(z) = z/r_\epsilon(z), \quad h_\epsilon(z) = r_\epsilon(z)R(u_\epsilon(z)). \quad (42)$$

For $\rho^2 = m^{-1}\|z\|_2^2$,

$$u_\epsilon(\lambda z) = \kappa(\lambda, z) u_\epsilon(z), \quad \kappa(\lambda, z) = \frac{\lambda \sqrt{\rho^2 + \epsilon_{\text{rms}}}}{\sqrt{\lambda^2 \rho^2 + \epsilon_{\text{rms}}}}. \quad (43)$$

Let L_R be a Lipschitz constant of the scalar curve over an interval containing all coordinates of $u_\epsilon(z)$ and $\kappa(\lambda, z) u_\epsilon(z)$. Applying this scalar Lipschitz bound coordinatewise, and then compensating the output matrix by λ^{-1} , gives

$$\left\| \lambda^{-1} h_\epsilon(\lambda z) - h_\epsilon(z) \right\|_2 \leq \left| \frac{r_\epsilon(\lambda z)}{\lambda} - r_\epsilon(z) \right| \|R(u_\epsilon(z))\|_2 \quad (44)$$

$$+ \frac{r_\epsilon(\lambda z)}{\lambda} L_R |\kappa(\lambda, z) - 1| \|u_\epsilon(z)\|_2. \quad (45)$$

Both terms vanish when $\epsilon_{\text{rms}} = 0$. With a bounded rescaling $|\log \lambda| \leq c$, the same bound is small when $e^{2c}\epsilon_{\text{rms}}/\rho^2$ is small and the global response and derivative are bounded on the compact interval above. Near the floor, or under an arbitrarily large reciprocal rescaling, the positive pair-rescaling identity no longer gives a useful invariance. MatrixPolicy therefore uses this direction only through a clipped log step with floored matrix RMS values and matching optimizer-state rescaling.

C MATRIXPOLICY UPDATE GEOMETRY

MatrixPolicy uses the RLB calculus in the order imposed by one optimizer step. The useful identities are gradient factorizations and log-ratio updates: the former separate the input and output matrix roles, and the latter explains why a reciprocal matrix rescaling is a bounded balance move. Back-propagation first produces gradients with respect to the pre-update parameters. Pressure and activity statistics are then computed from these gradients, A_l^t , B_l^t , and the rational coefficients before any parameter is changed. With EMA decay $\beta = 0.95$, the matrix and coefficient statistics have the form

$$q_{l,g}^{\text{in},t} = \beta q_{l,g}^{\text{in},t-1} + (1 - \beta) p_{l,g}^{\text{in},t}, \quad (46)$$

$$q_{l,g}^{\text{out},t} = \beta q_{l,g}^{\text{out},t-1} + (1 - \beta) p_{l,g}^{\text{out},t}, \quad (47)$$

$$q_{l,g}^{R,t} = \beta q_{l,g}^{R,t-1} + (1 - \beta) p_{l,g}^{R,t}. \quad (48)$$

Live curve gains are read from the RLB forward cache sampled by the optimizer’s configured statistics path and enter the group-gradient multipliers in equation 14. The Muon fraction in equation 24 and the balancing quantities in equations 28–29 use pre-update pressure/activity EMAs. Non-matrix Transformer parameters and rational coefficients follow AdamW. The RLB matrix blocks use the scaled gradients in equation 19 for the sequential AdamW and Muon matrix branches in equation 25. On scheduled rescaling steps, the positive matrix-pair move in equation 32 is applied after the matrix branch, and first- and second-moment-like optimizer-state buffers are scaled in the matching direction.

The need for separate input and output matrix rules follows from the gradient factorization around the RLB group. For one training example, let $\bar{y}_l = \nabla_{y_l} \mathcal{L}$ be the upstream gradient, and let $J_{l,g} = \partial h_{l,g} / \partial z_{l,g}$ be the group Jacobian in equation 40. The adjacent matrix gradients factor as

$$\nabla_{B_{l,g}} \mathcal{L} = \bar{y}_l h_{l,g}^\top, \quad (49)$$

$$\nabla_{A_{l,g}} \mathcal{L} = (J_{l,g}^\top B_{l,g}^\top \bar{y}_l) x_l^\top. \quad (50)$$

The output matrix is coupled directly to the realized feature $h_{l,g}$, whereas the input matrix is coupled to the tangent and radial components of the RLB Jacobian before the upstream gradient reaches x_l . Minibatch and sequence gradients sum these factors over positions and examples. Response RMS summaries therefore align with the output role, and derivative RMS summaries align with the input role. They remain proxies because the full output feature contains the group radius $r(z)$, and the full input gradient depends on $B_{l,g}$, the upstream gradient, and the radial term in equation 40.

The pressure signal is a stable magnitude proxy for motion along the input/output balance direction. In the idealized unfloored rescaling direction, an infinitesimal increase of the input representative and reciprocal decrease of the output representative has the loss derivative

$$\Delta_g = \langle \nabla_{A_{l,g}} \mathcal{L}, A_{l,g} \rangle_F - \langle \nabla_{B_{l,g}} \mathcal{L}, B_{l,g} \rangle_F. \quad (51)$$

MatrixPolicy does not estimate Δ_g directly. Instead it uses RMS relative-gradient proxies. For same-shaped nonzero unfloored $W, \mathcal{G}$, if $W^+ = W - \eta \mathcal{G}$ and $\eta \|\mathcal{G}\|_F < c \|W\|_F$ for a fixed $c < 1$, then

$$\log \text{rms}(W^+) - \log \text{rms}(W) = -\eta \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} + O\left(\eta^2 \frac{\|\mathcal{G}\|_F^2}{\|W\|_F^2}\right), \quad (52)$$

and Cauchy–Schwarz implies

$$\left| \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} \right| \leq \frac{\|\mathcal{G}\|_F}{\|W\|_F} = \frac{\text{rms}(\mathcal{G})}{\text{rms}(W)}. \quad (53)$$

The implemented quantities p^{in} and p^{out} replace the unfloored ratio on the right-hand side by floored RMS proxies. The floor makes the signals stable near zero, but it also means they are no longer exact upper bounds on $\text{rms}(\mathcal{G})/\text{rms}(W)$ near the floor. Thus $\pi_{l,g} = \log q_{l,g}^{\text{in}} - \log q_{l,g}^{\text{out}}$ is an RMS-imbalance proxy: its sign identifies which role has the larger EMA of the floored proxy, and its clipped magnitude supplies the pressure term used by equation 16 and equation 28.

The gain and pressure rules in equations 15–16 are log-space redistribution rules: high derivative gain reduces the input-role multiplier for that group, high response gain reduces the output-role multiplier, and a positive $\pi_{l,g}$ reduces the input-role multiplier while increasing the output-role multiplier. Coefficient activity in equation 17 damps groups whose rational curve is changing faster than the adjacent matrix pair. These corrections would change the layer-wide step scale unless they were centered. Before clipping,

$$\hat{c}_{l,g,\sigma} = \frac{c_{l,g,\sigma}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}}, \quad \frac{1}{G} \sum_{g=1}^G \log \hat{c}_{l,g,\sigma} = 0. \quad (54)$$

The policy redistributes update scale across groups while preserving the role-wise mean log multiplier before clipping. The subsequent clip in equation 18 bounds noisy group statistics and can break exact centering.

The pair-rescale target is motivated by equalizing representative input/output role scales in the positive log coordinate. For the unfloored nonzero-matrix calculation, write

$$\gamma_{l,g} = \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}). \quad (55)$$

An active rescaling step applies $A_{l,g} \leftarrow e^{\ell_{l,g,t}} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_{l,g,t}} B_{l,g}$, hence

$$\gamma_{l,g}^+ = \gamma_{l,g} + 2\ell_{l,g,t}. \quad (56)$$

If clipping is ignored and $\lambda_{l,g,t} = s_{l,t} \chi_{l,g}^{\text{rs}}$ uses the schedule and activity factor from equations 29–30, then equation 31 gives

$$\gamma_{l,g}^+ = (1 - \lambda_{l,g,t}) \gamma_{l,g} + \lambda_{l,g,t} \tau_{l,g}. \quad (57)$$

Because $0 \leq s_{l,t} \leq 0.625$ and $\chi_{l,g}^{\text{rs}} \leq \exp(0.125)$ under the stated clips, $0 \leq \lambda_{l,g,t} < 1$ before log-step clipping. Thus this is an exact contraction only in the nonzero-matrix, unfloored-RMS, $\epsilon_{\text{rms}} = 0$, unclipped idealization. The implementation uses the floored proxy in equation 27; near the floor, the log-ratio interpretation and the contraction model are approximate.

In the representative model that ignores group-dependent orientation, radius, and upstream-gradient constants, reciprocal rescaling makes the input-side role scale vary as $\hat{d}^{\text{probe}} e^{-\gamma}$ and the output-side role scale vary as $\hat{o}^{\text{probe}} e^{\gamma}$. Equalizing these two log scales gives $\gamma^* = \frac{1}{2}(\log \hat{d}^{\text{probe}} - \log \hat{o}^{\text{probe}})$. MatrixPolicy adds a smaller clipped RMS-pressure bias:

$$\tau_{l,g} = \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} \right) + 0.25 \bar{\pi}_{l,g}, \quad \bar{\pi}_{l,g} = \text{clip}(\pi_{l,g}, -1.25, 1.25) \quad (58)$$

using fixed-grid probe gains from equation 26. Probe gains make the balance target depend on the current curve shape rather than the current sampled feature distribution, while the $\bar{\pi}$ term lets recent matrix-pressure imbalance shift the target within a bounded range.

The final pair-rescale operation follows the approximate representative geometry above. The AdamW/Muon branch in equation 25 updates the represented map through the scaled matrix gradients. The time window inside the Muon fraction in equation 24 is

$$w(\xi_t) = \text{smoothstep}(0.02, 0.12, \xi_t) [1 - \text{smoothstep}(0.20, 0.36, \xi_t)]. \quad (59)$$

Together with the role factor $\varrho_{\sigma}(l)$, penalty $\psi_{l,t}$, and clipping in the full $\mu_{l,\sigma,t}$ formula, this window confines the matrix-direction update to a fixed early interval. The penalty $\psi_{l,t}$ reduces this branch when RMS imbalance or coefficient-gradient activity is high. Muon enters as a bounded, state-dependent matrix-direction substep for the RLB matrix pair.